Annika Stein

Professional Experience

01/2024 – today **Postdoctoral Research Associate**, *Johannes Gutenberg-Universität*, Mainz.
Analysis: Data analysis with LHC Run 2 + Run 3 data, targeting precision measurements of top quark properties (energy asymmetry, differential cross sections) and search for top + ALP via calorimeter jets (exotics group).
Hardware: R&D for the High-Granularity Timing Detector (ATLAS Phase-II upgrade), design and testing of flexible PCB cables, production database development, module assembly and loading.
Service & leadership: ATLAS control room shifter (Run Control & Trigger central desk, Calorimeter & Forward desk), leadership role in top quark reconstruction (convener) and analysis contact.
Teaching: supervision of students (bachelor / master / PhD), teaching, tutorial coordination, responsible for exams (level: Master and Bachelor).

11/2021 – 12/2023 **Research Associate**, *RWTH Aachen University*, Aachen, PhD Candidate Particle Physics.
Development of robust deep learning algorithms for object identification (heavy-flavour jet tagging). Supervision of students (including a DAAD Rise Germany intern), teaching, coordination and leadership, responsible for exams, practical courses (level: Master and Bachelor).

2019 – 2021 **Scientific Assistant**, *RWTH Aachen University*, Aachen.
Tutor for Experimental Physics; Editing an online course on physics in cooperation with Springer Nature (B.Sc.)

2017 – 2019 **Student / Scientific Assistant**, *Johannes Gutenberg-Universität*, Mainz.
Tutor for Experimental Physics, Mathematical Methods, STEM Helpdesk (B.Sc.)

Education

11/2021 – 06/2024 **PhD (Dr. rer. nat.)**, *RWTH Aachen University*, Aachen, Experimental Particle Physics.
Thesis: "Novel jet flavour tagging algorithms exploiting adversarial deep learning techniques with efficient computing methods and preparation of open data for robustness studies"

10/2019 – 10/2021 **Master of Science RWTH**, *RWTH Aachen University*, Aachen, excellent (grade 1.2).
Physics, Focus of Study: Astroparticle Physics and Cosmology. Master thesis:
"Investigation of robustness of b-Tagging algorithms for the CMS Experiment", grade: 1.0 (best possible)

04/2016 – 01/2019 **Bachelor of Education**, *Johannes Gutenberg-Universität*, Mainz.
Physics, Mathematics, Educational Sciences, Bachelor thesis (Mathematics):
"Die Gruppentheorie des Zauberwürfels und des Skewbs", grade: 1.0 (best possible)

08/2008 – 03/2016 **Abitur (High School Diploma)**, *Nikolaus-von-Kues Gymnasium*, Bernkastel-Kues.
Examination in: Physics, Mathematics, Geography, English

Qualifications

Languages: German (first language), English (fluent), French (good), Italian (basics)

Computing: Programming, tools: Python (numpy, matplotlib, awkward-arrays, scipy, pandas, astropy); Deep Learning (PyTorch, scikit-learn); C++, experiment-specific software (Athena, CMSSW); shell-scripting; Git / GitHub / GitLab, CI, project management, documentation (markdown, mkdocs); web development (HTML, CSS, JavaScript, WordPress); database / API client development; LaTeX; High-Performance / High-Throughput Computing: SLURM, HTCondor; GIMP, Inkscape, video editing; Linux, MacOS/iOS, Windows; docker, singularity; CAD / technical drawings

Soft Skills: Teaching, supervision, management and leadership (eight years experience), public speaking, event organization

Extra courses: ATLAS Control Room Shifter Training; Statistical Methods; QCD and Monte Carlo techniques; Computer Science (data structures and algorithms, databases and information systems, introduction to programming)

Teaching

2024 – **Coordinator (JGU)**, *Exercise classes, exams, team leader, occasional substitute lecturer*, Experimental Physics 5b – Nuclear and Particle Physics (B.Sc.), Particle Detectors (M.Sc.), Experimental Physics 4 – Scales and Structures of Matter (M.Ed., B.Sc.).

2023 **Coordinator (RWTH)**, *Exercise classes, electronic examination concept, team leader*, Data Processing in Physics (B.Sc.).

2019 – 2022 **Teaching Assistant (RWTH)**, *Supervision & grading of students' exercises and exams, discussion of course content*, Laboratory Courses on Particle Physics and Astroparticle Physics (Master) – top-quark physics, Advanced Laboratory Course Physics (Bachelor) – Mößbauer-Effect, Experimental Physics (Bachelor).

2017 – 2019 **Tutor (JGU)**, *Weekly exercise classes & grading of exams, mentor for STEM students at the department's afternoon coaching service*, Experimental Physics, Mathematical Methods (Bachelor), "Lernwerkstatt".

Invited Lecture

2025 **ATLAS HGTD Expert Week Facilitator**, *ATLAS, CERN*, Geneva, Hands-on tutorial / lecture on HGTD Production Database.

2023 **CMS Data Analysis School Facilitator**, *CMS, CERN*, Geneva, Hands-on tutorial after my lecture: Flavour-Tagging Short Exercise. Responsible for the team, documentation, concept, supervision.

Awards / Scholarships / Funding

10/2025 **Poster Award – Committee's Selection**, *ATLAS Collaboration Week*.
For "Top quark-antiquark pair reconstruction in the lepton+jets channel with SpaNet on detector- and particle-level", Paestum, Italy.

04/2025 **2025 Breakthrough Prize in Fundamental Physics**.
The 2025 Breakthrough Prize in Fundamental Physics is awarded co-authors of publications based on CERN's Large Hadron Collider Run-2 data released between 2015 and July 15, 2024, at the experimental collaborations ATLAS, CMS, ALICE and LHCb.

04/2023 **Travel Grant**, *Förderbeitrag für die erfolgreiche Teilnahme an der 86. Jahrestagung und DPG-Frühjahrstagung der Sektion Materie und Kosmos vom 20. bis 24. März 2023 in Dresden*, Wilhelm und Else Heraeus-Stiftung.

04/2022 **Best Poster Award – First Place**, *Center for Data and Computing in Natural Sciences (CDCS)*.
Opening Symposium, For "Improving robustness of jet tagging algorithms with adversarial training", DESY, Hamburg.

01/2020 **Preis für herausragende Abschlussarbeiten**, *FB Physik, Mathematik und Informatik, Johannes Gutenberg-Universität Mainz*, For my Bachelor thesis "Die Gruppentheorie des Zauberwürfels und des Skewbs".

03/2016 **BEGYS - Begabtenförderung am Gymnasium mit Verkürzung der Schulzeit**, *Nikolaus-von-Kues Gymnasium Bernkastel-Kues*, Fast-track program for outstanding students (secondary school level), by recommendation.

Professional Activities

Organization & Service

2026 Co-Organization ATLAS Workshop (Mainz)

2025 – ATLAS Top Reconstruction Group Convener

2025 – ATLAS Control Room Shifter (Calorimeter and Forward desk, Run Control and Trigger desk)

2025 – ATLAS HGTD Production Database Contact (Module and Assembly Group)

2024 – ATLAS Exotics Group Analysis Contact

2024 – y-ATLAS Local Representative (Mainz)

2023 – Peer Review for: European Physics Journal C (EPJ C, Springer); Journal of Open Source Software (JOSS, Open Journals)

2022 – 2023 Co-Organization RTG Colloquium (RWTH Aachen)

2022 – 2023 CMS b-Tagging & Vertexing Reconstruction Contact (led LHC Run 3 update)

2022 Local Organization Team, FSP CMS Workshop 2022 (RWTH Aachen)

2022 Co-Organization RTG Annual Retreat 2022 (Wiehl)

2022 Local Organization Team, Selected Interviewee Higgs10 Outreach Event (Exhibition, RWTH Aachen)

Memberships

2024 – ATLAS Collaboration, CERN.

2020 – 2023 CMS Collaboration, CERN.

2021 – 2023 DFG Research Training Group "Physics of the Heaviest Particles at the LHC", GRK 2497, RWTH Aachen

2022 – Deutsche Physikalische Gesellschaft (DPG)

Volunteer Activities

2014 – 2025 **World Cube Association (WCA) and German Cube Association e.V. (GCA)**, *International Non-Profit Organisation*, Delegate, participated in the WCA Advisory Council, Hosting "Speedcubing" competitions, solving Rubik's cubes and other brainteasers quickly; organisation of tournaments and team lead, software contributor.

Publications (selection)

2024 Annika Stein. **Novel jet flavour tagging algorithms exploiting adversarial deep learning techniques with efficient computing methods and preparation of open data for robustness studies.** PhD Thesis. RWTH-2024-07840, https://doi.org/10.18154/RWTH-2024-07840

2023 CMS Collaboration. **A first look at early 2022 proton-proton collisions at $\sqrt{s} = 13.6$ TeV for heavy-flavor jet tagging.** CMS DP-2023/012, https://cds.cern.ch/record/2854698

2023 Annika Stein. **Improving robustness of jet tagging algorithms with adversarial training: exploring the loss surface.** *21st International Workshop on Advanced Computing and Analysis Techniques in Physics Research (ACAT) 2022*, Proceedings in review for Journal of Physics: Conference Series, in press. arXiv:2303.14511

2022 CMS Collaboration. **Adversarial training for b-tagging algorithms in CMS.** CMS DP-2022/049, https://cds.cern.ch/record/2839919

2022 Annika Stein et al. **Improving Robustness of Jet Tagging Algorithms with Adversarial Training.** https://doi.org/10.1007/s41781-022-00087-1, *Comput Softw Big Sci* (2022) 6: 15. arXiv:2203.13890

Talks / Presentations (selection)

2025 **Top quark-antiquark pair reconstruction in the lepton+jets channel with SpaNet on detector- and particle-level.** ATLAS Week 2025 (Paestum, Italy)

2025 **Development of the Production Database of the High-Granularity Timing Detector for the ATLAS Phase-II Upgrade.** DPG Spring Meeting 2025 (Göttingen)

2024 **Novel Jet Flavour Tagging Algorithms exploiting Adversarial Deep Learning Techniques.** ETAP Seminar (Mainz)

2023 **Robust Jet Tagging Algorithms and AI Safety with CMS Open Data.** Invited talk for prolongation of the German Research Foundation (DFG) graduate school "Physics of the Heaviest Particles at the LHC" (Aachen)

2023 **Development of novel machine learning algorithms for robust jet flavour classification for Run3 at CMS.** DPG Spring Meeting 2023 (Dresden)

2023 **Direct search for Higgs boson decay to a pair of charm quarks in the vector boson associated production mode at CMS.** DPG Spring Meeting 2023 (Dresden)

2022 **Adversarial training for b-tagging algorithms in CMS.** (For the CMS Collaboration), ML4Jets Workshop 2022 (Rutgers University, US / online)

2022 **Improving robustness of jet tagging algorithms with adversarial training.** ACAT 2022 (Bari, Italy / online)

2022 **Improving robustness of jet tagging algorithms with adversarial training.** CDCS Opening Symposium 2022 (DESY, Hamburg)

Full list available online. https://annikastein.github.io/anstein/talks_theses.html (about 60 contributions, some internal)

Morbach, 1. Januar 2026